VIRTUAL 2315 CARTRIDGE FACILITY

Virtual 2315 Cartridge Facility

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Overview

The Virtual 2315 Cartridge Facility (V2315CF) is an enhancement to an IBM 1130 System which makes use of modern files as the contents of the disk cartridge, rather than having the disk drive read and write a physical 2315 cartridge.

The major objective of using V2315CF is protection of the extremely rare and sensitive disk heads in the internal disk drive of the 1130. The heads will no longer be loaded onto a platter inside a 2315 cartridge, thus they and the cartridge are safe from head crashes and other damage.

A small cylindrical object contains a file on an internal microSD card. This Mini 2315 Cartridge is inserted into the model disk drive on the V2315CF to allow the IBM 1130 to read and write to the Mini 2315 Cartridge as if it were a physical 2315 cartridge. The operator inserts different Mini 2315 Cartridges just as they would have inserted real 2315 cartridges in the disk drive.

The cartridge image file in the Mini 2315 Cartridge are similar to those used by various IBM 1130 Simulator programs, requiring use of conversion utility programs to switch between the V2315CF and simulator formats.

Thus, many cartridge images can be created and manipulated on personal computers, as well as making use of images that were archived from physical 2315 cartridges that had been used with IBM 1130 systems.

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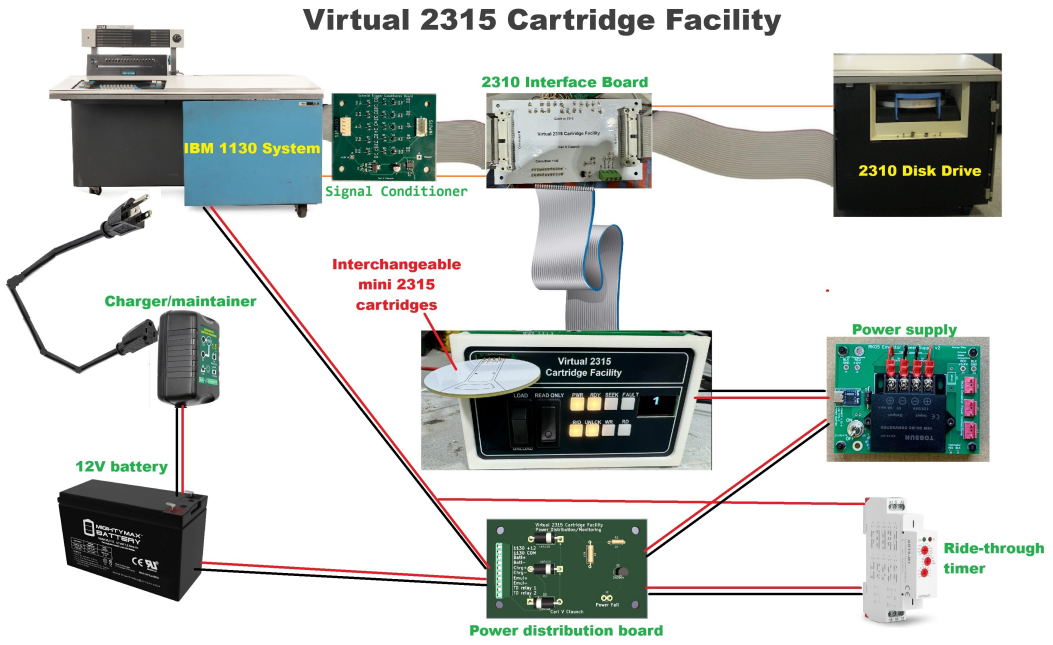
V2315CF implements a small model of the internal disk drive as seen from the front of the IBM 1130 when the front cabinet door is swung open. There is a blue handle that is pulled down to allow a 2315 Disk Cartridge to be inserted into or removed from the drive. The handle is returned to the up position and the toggle switch below it is flipped up to allow the mini cartridge to be used.

The user in front of the 1130 system enjoys the traditional experiences of using the internal disk drive; the whir of the cartridge spinning, the grunt of the arm moving the disk heads, the vibration of the motors, even the process of sliding a cartridge into the drive slot. The cartridge in the real inernal disk drive spins but is otherwise unused.

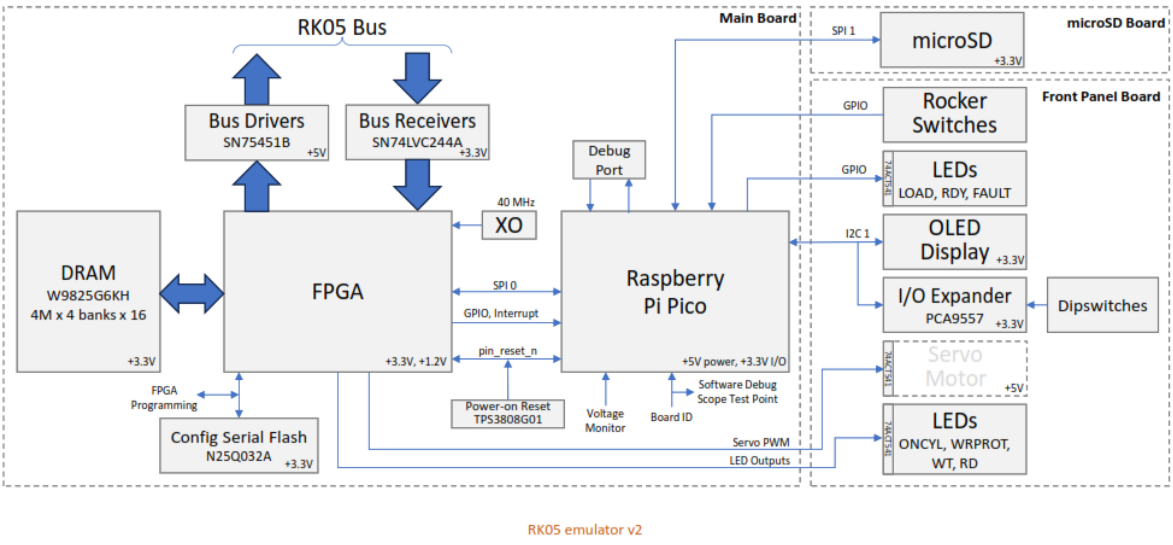
A secondary mode of operation is available - virtual - for use when the disk drive in the 1130 is not working. The 1130 system still sees a disk drive and can read and write to the cartridge images. All that differs in virtual mode is the lack of physical sensations from a drive’s operation as the drive is not turned on.

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Block Diagram of the V2315CF



The central box in the diagram is the main unit of the V2315CF. Mini 2315 cartridges are inserted into the opening of the mini disk drive on the right of the main box.



RK05 emulator v2

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The 2310 Interface Board routes signals between the IBM 1130 System, the V2315CF main box and the 1130's internal disk drive. It also contains the Real/Virtual mode switch and circuitry to monitor and control the Unlock lamp on the IBM 1130 console.

A signal conditioner PCB sits inside the 1130 to clean up signals coming from the V2315CF to the 1130 logic circuits. This deals with five signals that are produced by the V2315CF rather than the disk drive and whose signal quality issues would otherwise lead to erroneous behavior.

This board converts those poor quality signals into clean signals with a single on and off edge which will be correctly interpreted by the 1130's Solid Logic Techology (SLT) gates. The PCB is mounted to swing out gate A inside the 1130 near the hinge, where the twisted wire cable enters the 1130 coming from the 2310 Interface Board.

The internal disk drive grounds a wire to the V2315CF when the drive has unlocked the handle allowing someone to insert/remove 2315 cartridges. The wire is connected to the 2310 Interface Board and another wire from that board is connected to the lamp on the 1130 console. The lamp is controlled either by the disk drive or V2315CF.

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The power system of the V2315CF has a battery which is intelligently trickle charged whenever the IBM 1130 System is physically connected to building power, even if the 1130 system is switched off. The IBM 1130 system, when powered on, produces 12V power which is connected to the power supply of the V2315CF.

If the 1130 is not connected to the building power for many days or longer, the battery should be disconnected otherwise the trickle charger will slowly drain the battery leading to issues when trying to power up after the long period of downtime.

The 1130 supply normally powers the main unit and the 2310 Interface Board. The battery and 1130 system power are both connected to a Power Distribution Board, which in turn passes the power to the 2310 Interface Board and V2315CF main unit.

When the IBM 1130 system power fails, it is detected and a signal sent to the V2315CF which forces an unload of the disk image back to the Mini 2315 Cartridge. A timer keeps the battery connected to the main unit for about 45 seconds, long enough for the unload, then drops power until the 1130 system power returns.

If the disk drive is set to read-only mode, however, then even when power is lost, the V2315CF will NOT write the changed contents back to the mini cartridge.

IBM 1130 and its disk drive

The IBM 1130 System, first shipping in 1965, had a disk drive built into the machine. When the front cover of the 1130 was opened, the operator could pull on a handle to open a slot, into which a 2315 disk cartridge is inserted.

These removable cartridges each held 512K words of storage (1MB), organized in 203 cylinders. On each cylinder, four sectors of 321 words were available. As the drive had two heads, one for each side of the disk platter inside the cartridge, each cylinder held 8 sectors of data.

IBM had developed disk drives in the 1950s and early 1960s based around a 14” disk platter, coated with magnetic oxide. Their S/360 line of mainframes, also first delivered in 1965, used stacks of multiple 14” platters as disk packs. However, the smaller IBM 1130 used a single platter in a cartridge - the 2315. The disk drive that used the 2315 cartridges was internally named the 13SD. Within the IBM 1130 documentation it is generally just referred to as the ‘internal drive’.

The drive was also available in a standalone box called the 2310; the box could hold 1 or 2 of the 13SD drives. The 2310 could also be attached to other systems such as the IBM 1800, a cousin of the 1130.

I will use 13SD, 2310 and internal drive interchangeably for the disk drive inside the IBM 1130 cabinet in all the documents and designs related to the V2315CF.

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Mini 2315 Cartridge as an interchangeable file holder

The disk image used by the V2315CF is held on a microSD card. That file is read in and loaded into RAM inside the V2315CF when the switch on the main unit is set to Load.

All reads and writes by the IBM 1130 take place to and from the RAM. After a session using the cartrdige it is likely that many sectors were written with changed or new content. These must get back onto the slower microSD card when the session ends.

When the main unit is set to Unload, the data from RAM is fetched and the disk image on the microSD card is rewritten. Changes made by the IBM 1130 dring the session are preserved and seen when the Mini 2315 Cartridge is next loaded.

Optionally, by toggling the Read-Only switch during the session, one can skip the rewrite of the image when unloading. This allows Mini 2315 Cartridges to be kept in their original unaltered form, with all updates during a session discarded at the end.

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A microSD card is a small object, one that is also somewhat fragile. Thus, I created a holder PCB as a 2” diameter replica of a 2315, containing a microSD card plugged into its connector. This PCB is in turn inserted into the bottom of the mini 2315 cartridge, about 3 3/4” in diameter, modeled to look like a smaller version of the 2315.







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It is envisioned that users of the V2315CF will keep a number of virtual 2315 cartridges, each one embodied in a Mini 2315 Cartridge, interchanging them as desired. These slide into a small model of the internal disk drive front panel.

The user must insert a mini 2315 cartridge into the model disk drive, flip on the tiny motor switch below it, then flip the rocker switch on the V2315CF front panel to LOAD in order to have the virtual 2315 cartridge loaded into the drive before one attempts to turn on the physical drive and spin up the motor.

The V2315CF is mostly housed in a black plexiglas rectangular solid enclosure which is installed on top of the internal disk drive, accessible by opening the top lid of the IBM 1130. The main panel of the V2315CF is on the left of the faceplate and the model disk drive into which the mini cartridge slides is on the right of the faceplate.

Cables from this enclosure to the 2310 Interface Board that controls signals flowing between the itnernal disk drive and the 1130's disk controller logic. In addition to the 2310 Interface Board and main enclosure, there is a smart/trickle charger and a 12V motorcycle style battery plus wires bringing the 1130's 12V power rail to the V2315CF.

The smart/trickle charger is plugged into the utility outlet on the right side of the IBM 1130, one that is always powered when the main power cable of the 1130 is plugged into the building electrical supply. The battery is secured to the bottom of the 1130 underneath the disk drive.

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Using the V2315CF with the IBM 1130

Real mode operation

Real mode is a hybrid operation of the internal disk drive in the IBM 1130 working in conjunction with the V2315CF. It is the normal method of operation. This is the normall mode.

Power up the IBM 1130 system, if it is not already running. Ensure that the internal disk drive of the IBM 1130 has its motor switch (below the blue slot handle) turned off. The UNLOCK lamp on the 1130 console will be lit, as well as the UNLK lamp on the front panel of the V2315CF indicating that you can insert or remove mini 2315 cartridges.

A supplied physical 2315 cartridge is inserted into the internal disk drive of the IBM 1130 and the blue slot handle closed. The 2315 cartridge is an eight sector type such as used with the 1130 but the contents or condition of the platter doesn’t matter.

The V2315CF main unit begins with its Load/Unload switch set to Unload. A Mini 2315 Cartridge is inserted into the model disk drive on the front of the V2315CF and its tiny motor switch turned on.

Turn the Load/Unload switch on the front panel to Load. The display screen will show the progress reading the virtual 2315 cartridge contents and finish with a 2315 icon on the screen.

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Turn the 1130 internal disk drive motor switch on (up position). You will hear the disk cartridge begin to spin and after about 90 seconds the drive should reach its ready condition where it normally would have loaded the disk heads onto the platter.

The RDY lamp on the V2315CF will light, in addition to the green File Ready lamp illuminating on the console of the IBM 1130. That lamp is to the left of the keyboard,. This indicates that the system is ready for software to make use of the disk drive.

Use the disk drive normally, as if it were just the internal disk drive in its unaltered state. Software issuing seek commands will move the disk arm in and out among the 203 cylinders (radial positions) on the platter. Read or write individual sectors.

The internal disk drive will move its disk arm to obey the seek commands issued by software, providing a grunting sound and some vibrations as the arm moves. The SEEK lamp on the V2315CF will flash briefly for each seek that is performed.

The V2315CF is also tracking the current cylinder for the virtual 2315 cartridge that was loaded, so that any read or write will occur at the current cylinder location of the heads

When the software reads a sector into core memory on the IBM 1130, the RD lamp will flash on the V2315CF. When software writes a sector from core memory, the WR lamp flashes. The data is moving between core memory and the V2315CF main unit, not the internal disk drive.

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Running the DMS2 operating software of the IBM 1130 is done the same way as if the internal disk drive was unaltered. You load a Mini 2315 Cartridge which has DMS2 installed on it. The V2315CF loads the cartridge image to RAM and the disk drive lights the File Ready and RDY lamps.

A boot card is placed in the primary input device, usually a 1442 or 2501 card reader, and the Prog Load button on the IBM 1130 console is pressed. The card is read into memory and that card will read DMS2 from the disk drive and begin its execution.

When it is time to stop using the disk, or you wish to change cartridges, turn off the internal disk drive motor switch (down). You will see the File Ready and RDY lamps go out, hear the spinning cartridge slow down, and eventually the Unlock lamp on the IBM 1130 console, near the File Ready lamp, will illuminate. In addition, the UNLK lamp on the V2315CF turns on.

As the drive spins down, turning the Load/Unload switch on the V2315CF to Unload causes the contents of the disk as it has been changed during your session to be written back to the Mini 2315 Cartridge.

The OLED screen on the V2315CF will display the cylinders being written back and then end with the idle screen image. The Mini 2315 Cartridge can be removed from the model disk drive on the front panel of the V2315CF.

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If a new cartridge is desired during any session, after having run using one cartridge and unloaded it, insert a different Mini 2315 Cartridge into the model drive and flip the Load/Unload switch back to Load. The new cartridge will load. The internal disk drive motor switch should be turned on, then after about 90 seconds the File Ready and RDY lamps light to indicate that the new cartridge is ready for access.

As an alternative to updating the Mini 2315 Cartridge with the changes made to the cartridge during a session, the V2315CF may be put into read-only mode. This is done by toggling the Read Only switch to set the RO lamp on.

If RO is lit when the Load/Unload switch is turned to Unload, the V2315CF will skip writing back the changes to the Mini 2315 Cartridge. The OLED displays the idle screen.

Thus the contents of that Mini 2315 Cartridge are the same as when it was first loaded - any changes made to the cartridge during your session are lost.

This is useful if one wants a training or student version of DMS to be available to multiple people in the same state in which it was prepared.

Each student or museum visitor can use DMS2, compiling and storing programs, updating files and experimenting, seeing those changes during their session, yet the next visitor or student will have the same initial setup because the Mini 2315 Cartridge did not change.

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Virtual mode

Virtual mode is an alternative that does not use the 1130 internal disk drive. That drive does not have its motor switched on and the disk arm will not be moving in concert with the V2315CF.

There is a tiny slide switch on the 2310 Interface Board inside the IBM 1130 on the right side. Slide it to the virtual setting before powering up the 1130. Leave the motor switch (below the blue slot handle) of the internal disk drive off. The UNLOCK lamp on the 1130 console panel and the UNLK lamp of the V2315CF should be lit.

The V2315CF main unit starts with its Load/Unload switch set to Unload. A Mini 2315 Cartridge is inserted into the model disk drive on the front panel and the tiny ‘motor’ switch on the model drive flipped up.

Turn the Load/Unload switch on the V2315CF to Load. The OLED display screen should show the progress reading the virtual 2315 cartridge contents and finish with a 2315 icon shown on the screen. Remember, the internal disk drive motor switch is not turned on when running in virtual mode.

After 90 seconds, you will see the RDY lamp on the V2315CF light, in addition to illuminating the green File Ready lamp on the console of the IBM 1130, to the left of the keyboard. This indicates that the system is available for software to make use of the disk drive.

Use the disk drive normally, as if it were just the internal disk drive in its unaltered state and the same way you would use V2315CF in Real mode.

When ready to end a session and stop using the disk, flip the Load/Unload switch of the V2315CF to Unload. It will then turn off the File Ready and RDY lamps, turn on the UNLK lamp, and rewrite the changed contents back to the Mini 2315 Cartridge. The OLED will return to its idle screen.

If the RO lamp is on, then the changed contents are not written back, in the same way that read-only mode works with in Real mode.

Use of the IBM 1130 disk drive with the V2315CF in Virtual mode is the same as with Real mode, except you will not hear a cartridge spinning, hear grunts of the arm moving, nor feel the vibrations of the drive. The software uses the disk with exactly the same performance as in Real mode.

Selecting Real or Virtual mode

A small switch on the 2310 Interface Board, located inside the right side cover of the IBM 1130, is moved between Real (left) and Virtual (right) positions to control the mode. It should be moved while the 1130 is powered off.

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V2315CF is a derivative of an RK-05 Emulator

George Wiley created the RK-05 Emulator, which emulates DEC disk drives and serves up cartridge images from microSD cards. Because the RK-05 drive is derived from the IBM 13SD technology, under license, the emulator was a good starting point to develop the V2315CF.

However, very substantial differences exist between the 13SD and RK-05, the disk formats of the cartridges, and the disk controller electronics in the computing systems. The RK-05 emulation has the operator directly handle and insert the microSD cards for each RK-05 cartridge image into a the microSD socket.

The RK-05 Emulator has a Raspberry Pi PICO processor, an FPGA, SDRAM, circuits to interface with the electronics of a DEC controller, controls, LEDs and other components.

Interface cards provide twin ribbon cables from the RK-05 emulator to the disk controller in the DEC system. A terminator board ensures that the signal lines are properly terminated to ensure good signal quality.

Fortunately, the number of input and output signal lines for the RK-05 Emulator are sufficient to meet the needs of a 13SD connected to an IBM 1130. I therefore leveraged the main box circuitry but reprogrammed both the FPGA and PICO to my needs.

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I substituted a new faceplate on the main box and modified the method of connected microSD cards. The RK-05 Emulator has a small PCB with the connector for the microSD card sitting in a slot, with a 2x5 ribbon cable connecting that PCB to the remainder of the emulator.

I relocated the ribbon cable end to the model of the disk drive that sits on the front panel of V2315CF to the right of the controls. My Mini 2315 Cartridge objects have a 2x5 pin connector on the back edge which plugs into the ribbon cable connector.

The objects are inside a realistic miniature of a 2315 cartridge with the pins sitcking out of the back. Sliding the mini cartridge into the model drive engages the pins in the ribbon cable connector and allows the V2315CF to access the virtual disk image that is stored on a microSD card inside.

I retained the power supply, interface cable board and terminator unmodified. I changed the meaning of some of the LEDs on the main unit, reflected in the labels on my replacement faceplate.

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Adapter to access virtual disk images on a mini cartridge

An adapter box is supplied with the V2315CF that has a USB cable coming from it an a female 2x4 connector on the face. A mini cartridge is plugged into the adapter and then the USB cable is plugged into a personal computer. The contents of the microSD card inside are opened on the PC as another drive.

Virtual disk images, files with a suffix of .dsk, are what embodies a virtual 2315. One such file can exist per mini cartridge. The adapter allows the file to be extracted or replaced from the PC when desired.

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Signaling levels used with the V2315CF

The IBM 1130 Solid Logic Technology (SLT) generally uses +3V as logic high and 0V as logic low. SLT is quite tolerant of higher voltages for logic high and even negative voltages for logic low. The 1130 also uses +48V and 0V for a few signals.

The PICO processor in the V2315CF requires that its IO pins be limited to +3.3V for logic high and no less than 0 for logic low.

The FPGA inside the V2315CF is also operated with +3.3V and 0V IO pins.

Because the DEC controller logic and RK-05 disk drives operated with +5V logic high and 0V low, the V2315CF contains voltage translation circuitry such that all signals on the ribbon cables running to the 1130 controller circuity and to the 13SD disk drive operate at those levels.

The 2310 Interface Board implements some sensing and control circuits connected to the main V2315CF unit, thus these operate at +5 and 0V levels.

The Unlock signal from the 13SD drive and the Unlock lamp on the IBM 1130 console operate at +48 and 0V.

The power fail sense circuit on the Power Distribution Board senses the +12V supply but outputs with an open collector gate thus is compatible with 3.3 and 0V levels of the PICO.

Power System of the V2315CF

Normal power for the V2315CF is supplied by the 12V power rail of the IBM 1130 system, when the 1130 is powered on.

This is fed through the Power Distribution Board to the V2315CF Power Supply (PS) unit. That power supply unit converts the input to 5V power which drives the V2315CF main unit. Internally, the main unit converts the 5V down to 3.3V for use with the FPGA, PICO, microSD card reader, OLED and other components.

The 12V from the Power Distribution Board (PDB) is also connected to the 2310 Interface Board where it powers a few circuits on that board.

A 12V battery in the system is also connected to the V2315CF, so that it can power the main unit when power drops from the IBM 1130. The design of the IBM 1130 causes the 12V to be rapidly cut off when the 1130 shuts down.

A circuit on the PDU detects the loss of the 12V supply from the IBM 1130 and produces a signal to the V2315CF that power was lost. The V2315CF will do an immediate unload of any cartridge that was being used, writing its changed contents back to the Mini 2315 Cartridge.

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A ride-through timer is connected between the PDU and the V2315CF PS. It senses the 12V coming from the IBM 1130 power supply, but switches the output of the PDU, which is a combination of the 1130 supply and the battery.

The timer is configured to connect the PDU to the PS whenever it sees 12V from the IBM 1130, and to maintain the connection for about 40 seconds after the IBM 1130 12V supply drops out. This ensures adequate time for the V2315CF to perform the emergency unload to the Mini 2315 Cartridge.

The PDU is fed by both the 1130 12V power supply and the 12V battery, thus when the 1130 supply drops out the battery picks up the load. Because of the ride-through timer, the battery is not powering the V2315CF for more then 40 seconds when power to the IBM 1130 is turned off. A battery charger/maintainer is connected to the battery and will keep it fully charged.

The charger is plugged into the IBM 1130 utility outlet that always has power as long as the main power cord of the IBM 1130 is hooked to building power supply, whether or not the IBM 1130 is switched on.

Diodes on the PDU isolate the IBM 1130 12V and the battery from each other. The voltage of the IBM 1130 12V power rail is typically several volts higher, thus it provides the majority of the power to run the V2315CF even though the battery also feeds some power through its diode.

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Error messages and codes

When something goes wrong using the V2315CF, in most cases a message on the LED screen indicates the cause. A TTL level USB serial connector can be hooked to pins on the back of the V2315CF main unit and a serial terminal program can read debugging stream messages at 115,200 baud 8-N-1 on a K\PC or workstation.

The serial debugging stream logs quite a bit of information about the state and progress of the V2315CF. Typing L on the terminal program connected to the debugging stream will enable messages issued for each read, write or seek performed with the V2315CF.

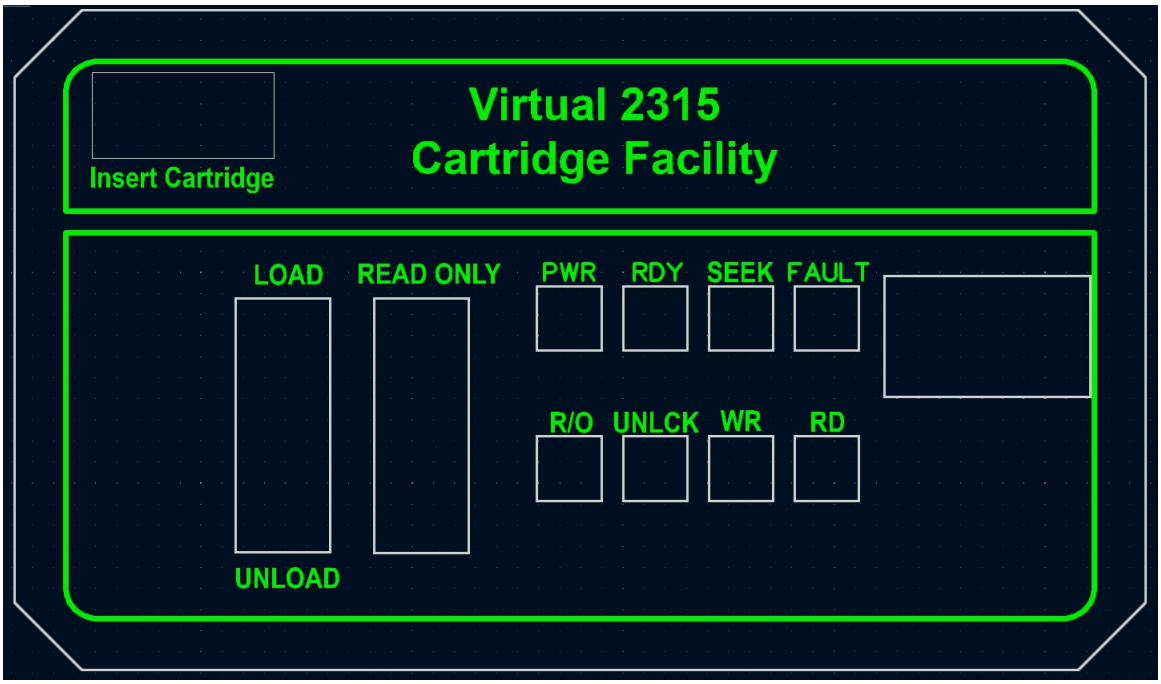
For example, if the 2315 mini-cartridge is not plugged into the model disk drive properly or the tiny motor switch on the model disk drive is not flipped up, when attempting to load it, the LED will state that the 2315 cartridge is not inserted and the FAULT indicator will flash 7 tiimes repeatedly.

The LED screen displays the progress reading the virtual 2315 image from the mini cartridge and loading it to the RAM inside the V2315CF - listing each ten cylinders loaded until all 203 have been loaded or rewritten.

Absence of errors is shown by a blank screen when no virtual 2315 cartridge is loaded or an icon of a 2315 with the cartridge number displayed across the bottom when it has been loaded.

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Front panel indicators and controls



The Load/Unload rocker switch is flipped to the up position to request that a mini cartridge be loaded into the V2315CF or flipped down to request that it be unloaded so that it can be removed from the model disk drive.

The Read Only rocker switch is a momentary action switch, pushed up to toggle between read-only and normal mode. The R/O lamp will light if the loaded mini cartridge will not be updated with any data written to the V2315CF from the 1130.

The right side LED screen displays status information. When a mini cartridge has been successfully loaded, an icon of a 2315 is displayed with the cartridge number, four hex digits, displayed along the bottom. Status and error messages also display here.

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The PWR lamp is lit when the V2315CF is powered up. This is on whenever the IBM 1130 system power is on as well as for about 40 seconds after power drops.

The RDY lamp is lit when the disk drive is in File Ready state, allowing software in the IBM 1130 to read, write and seek the drive arm of the virtual disk drive. This matches the FILE READY lamp on the front panel of the IBM 1130 which lights at the same time.

The UNLCK lamp is lit whenever the UNLOCK lamp on the main 1130 console is also lit. It is on when the internal disk drive of the 1130 has its motor off and the platter is not turning, thus allowing an operator to open the disk drive handle and withdraw or insert a 2315 disk cartridge. When the motor switch of the internal disk drive is turned on, with a cartridge inserted, the lamps go out.

This also signals when a mini 2315 cartridge can be inserted or removed from the model disk drive on the V2315CF front panel When in virtual mode, as soon as a mini cartridge is loaded using the Load/Unload rocker switch, the UNLOCK lamp and UNLCK indictor go off and after about 90 seconds, the FILE READY lamp and the RDY indicator should turn on.

In real mode, once the mini cartridge is loaded via the Load/Unload rocker switch, the motor swtich of the internal disk drive must be turned on and the motor spin up. After about 90 seconds the FILE READY lamp and RDY indicator turn on, indicating that the 1130 can access the disk.

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The RD, WR and SEEK indicators light for a brief time whenever the 1130 issues a command to read a sector, write a sector or move the disk arm to a new cylinder location. These allow the user to see that disk activity is occuring.

The FAULT indicator turns on when certain error conditions occur. This ight be due to an issue trying to read in the virtual 2315 image from the mini cartridge or to some error in the internal disk drive or other hardware.

The FAULT indicator flashes in a pattern to indicate the type of fault encounted for errors trying to load or unload a virtual 2315 cartridge image. The indicator flashes seven times for errors trying to load an image, three time for errors trying to update the mini cartridge with changes during an unloading.

This will remain flashing indefinitely until the user holds the Read-Only rocker switch depressed whle flipping the Load/Unload rocker switch up to Load. This resets the flashing and restores normal operation once the cause of the fault is found and rectified.

Loading errors include:

- no mini cartridge inserted or mini motor switch not up
- can’t access microSD card file system
- Error opening a file with a .dsk suffix
- bad format or data on the file
- can’t close the file after reading it

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Utility programs and tool for handling cartridges

Disk image files used by the simh based IBM 1130 simulator programs are not in the correct format to use with the V2315CF. The V2315CF format includes information about the cartridge identifier and other data provided by the user. Files need to be converted using a utility program that runs under Python on a Windows PC.

convert1130.py is a program that asks you to select the simulator format disk file, then select a location and name for the V2315CF format disk file. It will ask you to provide a cartridge identifier and a text description of what this cartridge is for.

convert2315.py is a program that will take a V2315CF format disk file and convert it to the format needed to run it on an IBM 1130 simulator. You choose the V2315CF format file and then select a location and name for the simulator format output. It lists the cartridge ID and other information from the V2315CF format file header.

listcartridges.py is a program that asks you to select a folder/directory, then lists all valid V2315CF format disk files located in that folder. It provides the file name, the cartridge identifier and the description from the file header.

Two programs are available to look inside or update a V2315CF format disk file. One will display the 321 words in hex for a specific sector The other will let the user edit a specific sector then update the disk file.

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showsector.py will list a sector from a V2315CF cartridge image file in hex words. You are asked to select the V2315CF format file, then enter a cylinder, head and sector number to be listed. Once the sector is listed, it offers the choice to select another sectors or quite the program.

updatesector.py will let the user edit a chosen sector using a graphical editor. First you are asked to select the V2315CF format file, then enter a cylinder, head and sector number to be edited.

The current contents of the sector are displayed in a grid, in hex format. Clicking in any cell allows entry ot a valid four character hex value that will replace that word in the sector. When the grid is closed, the updated sector is written to the file.

Theory of Operation

The heart of the V2315CF is a PCB that contains an FPGA, SDRAM and a Raspberry Pi Pico. The FPGA interfaces with the disk drive and IBM 1130 disk controller electronics. The Pico provides the user interface and manages the microSD card that holds the virtual disk image.

The FPGA and the Pico communicate over a Serial Protocol Interface (SPI), exchanging status and commands. The SDRAM is directly accessed by the FPGA.

At startup, the code in the Pico initializes the general purpose IO pins on the Pico, starts the SPI link with the FPGA, and initializes the FPGA. It also controls a small OLED screen that displays some status messages during operation.

When the operator inserts a Mini 2315 Cartridge, they are connecting a PCB holding a microSD card. SPI communications are used by the Pico to read and write to the microSD card, when the operator loads or unloads a virtual cartridge.

Loading a virtual cartridge begins when the Load/Unload switch on the V2315CF is turned to the Load position. The Pico opens the microSD card and looks for a file with the suffix .dsk which will be the disk image. The file is checked to be certain it is the correct format, then read by the Pico.

Virtual 2315 Cartridge Facility

The Pico writes the data from the microSD card file through the SPI link to the FPGA. The FPGA writes the data into SDRAM so that it is ready for disk operations by the IBM 1130.

The Pico informs the FPGA that a cartridge image was loaded into the SDRAM. The Pico then waits for the disk drive to become ready.

In real mode, the disk drive in the IBM 1130 will assert the File Ready signal line after it has been spinning for 90 seconds and believes all is ready for use. Thus the operator must have switched on the disk drive in the 1130 to spin the cartridge.

The V2315CF raises the File Ready signal to the 1130, when the cartridge image is loaded to RAM and the disk drive is ready which causes the File Ready lamp on the 1130 console to light up.

In virtual mode, the V2315CF begins to immediately spin up the virtual disk drive when a cartridge image is loaded. After a wait of 90 seconds File Ready is asserted to the IBM 1130 causing the File Ready lamp to light.

The Pico periodically checks to see that the disk is still ready for access - asking the FPGA over the SPI link for a status word, one of whose bits means the disk is ready.

If the ready bit turns off, such as when the operator switches off the disk drive in the 1130, the Pico turns off the RDY light and gets ready to retrieve the updated contents of SDRAM so that the file on the microSD card can be kept current.

Virtual 2315 Cartridge Facility

This is done by fetching RAM from the FPGA over SPI and writing the file back to the microSD card, unless we set Read Only mode.

When the 1130's disk drive is not turning, the Unlock lamp on the 1130 console is illuminated, as well as the UNLCK lamp on the V2315CF. When the motor starts spinning, the lamp goes off. In virtual mode, the lamp goes off as soon as we load a cartridge image and turns back on when the cartridge is unloaded.

While the disk is ready, the V2315CF passes commands between the 1130 and its disk drive, passes status back from the drive to the 1130, except for the data being read or written.

Thus, when a program issues a XIO Control to seek the disk arm some number of cylinders forward or backward, V2315CF passes that to the 1130 drive which physically performs the arm movement. The V2315CF is also monitoring the commands so that it knows the current cylinder of the disk drive.

The physical 2315 cartridge in the 1130's disk drive has notches cut into a ring on the bottom of the platter, which cause sector and index marker pulses to be sent from the drive to the V2315CF. In real mode, these are passed on to the 1130. In virtual mode, we simulate a spinning platter and generate the sector and index pulses to send to the 1130.

Virtual 2315 Cartridge Facility

When a program executes an XIO Initiate Read, it specifies a sector number (0 to 3) and a head (top or bottom) where we will read a sector. The buffer area in core memory has a count in the first word that tells the 1130 how many words to read from the sector. A sector can contain up to 321 words.

When the 1130 sees the sector pulse that marks the start of the targetted sector, it raises a Read Gate that normally would cause the disk drive to begin streaming the data from the disk head to the 1130.

The disk drive of the 1130 will take the serial stream of pulses and separate them into clock and data pulses. Every bit on the disk drive spans two pulse intervals. The first always has a pulse called the clock. The second interval only has a pulse if the bit value is 1, otherwise there is no pulse.

The 1130 drive separates the stream so that the clock pulses come on the Read Clock wire and the data pulses (for bit values of 1) come on the Read Data wire. Whether we are in real or virtual mode, we ignore the Read Clock and Read Data lines from the drive.

That is because we mechanically altered the drive so the heads do NOT ride down on the platter surface inside the cartridge. They remain safely above the disk surface and thus safe from disk crashes.

Virtual 2315 Cartridge Facility

The V2315CF starts timing from the sector pulse when the Read Gate is turned on. We produce the stream of bits that would come from the disk head, sending them as Read Clock and Read Data to the 1130. This includes the special bits - a preamble of zero bits, a sync word, and four check bits at the end of every data word in the sector. V2315CF generates the appropriate check bits based on the word we fetched from SDRAM.

The 1130 takes this stream from the V2315CF and handles them the same as when the 1130 disk is working as it was designed. The words from the sector are assembled, error checking is done on the four extra bits, and the data word is stored in core memory in the buffer.

When a program issues an XIO Initiate Write, it specifies a head and a sector number. The buffer for the write has a word count in the first word and then the data to be written to disk in the target sector.

The disk drive produces a clock that drives the writing of pulses onto the disk surface. The 1130 sets up a Write Clock and Data signal line to indicate whether to write a pulse or not at the next clock transition. The V2315CF passes the drive's clock to the 1130 in real mode; in virtual mode we generate the clock in the V2315CF and send it to the 1130.

The 1130 is always requesting a pulse for the clock interval but uses the data bit value to request a pulse or not in the data interval. Thus writing involves a single serial stream from the 1130 to the drive.

Virtual 2315 Cartridge Facility

The 1130 produces the string of zeroes (preamble), the sync word pattern, then sends each word in turn while appending four check bits after the word. The V2315CF captures the value of the Write Clock and Data line at the time of each oscillator transistion, breaks it into clock and data intervals, the assembles the words being written.

The V2315CF checks the four error bits but discards them otherwise. We write each word as it is received into the SDRAM, overwriting the previous contents of the sector.

In real mode, when the operator switches off the disk drive in the 1130, the drive drops File Ready and the FPGA detects this. The Pico asks for status periodically, sees the disk is not ready, and thus prepares to unload the cartridge.

Once the physical drive has come to a complete stop, the drive turns on the Unlock line and we light the Unlock lamp on the 1130 console and UNLCK light on the V2315CF.

After this point, when the operator switches the Load/Unload switch of the V2315CF to Unload, the contents of SDRAM are requested by the Pico, over SPI, to the FPGA which reads them and sends them over the link.

The Pico will rewrite each sector of data in the file on the microSD card, thus upon completion of the unload our file contains all changes made by the 1130 during use of the disk drive.

Virtual 2315 Cartridge Facility

In virtual mode, the disk drive is not turning. The operator flips the Load/Unload switch to Unload to cause the V2315CF to drop File Ready. The Pico will request the contents of SDRAM from the FPGA over SPI, just as in real mode, then rewrite the virtual cartridge file on microSD card.

The FPGA will light the SEEK, RD or WR lamps on the V2315CF for about a quarter of a second each time the 1130 does a seek, sector read or sector write. This is long enough for human perception, although a sector read or write is complete in less than 10 milliseconds.

Thus these lamps gives the operator a sense of the disk activity. In real mode, the operator will also hear the grunting sound of the 1130 disk drive moving the arm back and forth.

The Unlock lamp on the IBM 1130 console is controlled by the internal disk drive when in real mode, connecting a switch in the drive to the lamp through a relay on the 2310 Interface Board. When in virtual mode, the logic in the V2315CF will control the lamp using a MOSFET transistor on the 2310 Interface Board.

The V2315CF has a Read Only momentary toggle switch which is used to set or clear Read Only mode. The RO lamp on the V2315CF lights when in Read Only mode.

Read Only will skip fetching data from SDRAM and rewriting the cartridge image file when we are unloading a cartridge. This has the effect of reverting the image on the Mini 2315 Cartridge to the same state it was in when we started, regardless of what was written to or changed during our session using the 1130.

Virtual 2315 Cartridge Facility

During a session, even with RO lit, every XIO Initiate Write changes the sector being targeted and later XIO Initiate Read will pull in the altered data from the sector. Compiling programs, updating libraries and accessing data on the disk works completely normally while the disk is operating.
The cartridge image is rolled back, in effect, to the contents it had when the cartridge was loaded. This is handy to support museum visitors running work that might impact other visitors or museum demonstrations - their session is lost when the cartridge is unloaded in Read Only mode.

IBM 1130 systems provide multiple voltage rails, among them 12V and 48V. These latter two are only connected via a relay when the logic voltages such as +3, +6 and -3V are stable. When the power switch of the 1130 is turned off, the relay drops out thus rapidly shutting down the 12V supply.

The V2315CF Power Distribution Board is fed 12V from the IBM 1130, which is the primary source of power for operation. In addition, a 12V battery is connected to the Power Distribution Board so that the V2315CF has time to safely fetch SDRAM contents and rewrite a cartridge file if one was loaded at the time of power loss.

A monitoring circuit on the Power Distribution Board notifies the Pico when the 12V from the 1130 has dropped. The Pico continues to run, being powered by the 12V battery, while the code does an emergency unload of any cartridge image that was active.

Virtual 2315 Cartridge Facility

A ride-through time begins counting when the 1130's 12V supply drops. A timer activates when that happens. It switches the input of the V2315CF Power Supply module from the 1130's 12V power to the battery when activated for about 40 seconds, time enough for the emergency unload.

The battery is connected to an intelligent charger/maintainer that will keep it topped up. This charger is fed 120V from the utility outlet inside the 1130. As long as the main power cord of the 1130 is connected to building power, the charger works even if the 1130 is switched off.

The V2315CF Power Supply module converts the 12V coming from the timer switch into 5V that is fed to the main V2315CF box.

The 12V that is switched by the ride-through timer is also hooked to the 2310 Interface Board, powering its circuitry. The interface board receives power while the 1130 is up and for about 40 seconds after power from the 1130 ends, same as the Power Supply Module of the V2315CF.

Each Mini 2315 Cartridge has a PCB with a microSD card connector and a few other components. The microSD card with the virtual disk image sits inside a plastic mini 2315 cartridge which is inserted into a model disk drive on the front panel.

Pins on the back edge of the Mini 2315 Cartridge mate with a female connector in the rear of the model disk drive, connecting it to the SPI link which the Pico uses to read and write the file on the card.

Virtual 2315 Cartridge Facility

It is expected that an owner will have multiple Mini 2315 Cartridge units, much as they would have had multiple physical 2315 cartridges when using the disk drive as originally designed.

The physical 2315 cartridge that is inserted into the 1130's disk drive is only used to produce the sector and index marker pulses as well as provide the vibration and noise typical of the drive.

Thus a single physical 2315 cartridge remains in the 1130 internal drive but many different Mini 2315 Cartridges can be swapped and used with the 1130.

Disk images can be created or updated on the IBM 1130 simulators, such as the Brian Knittel's from ibm1130.org or Claudio Vincenzi's simulator.

Utility programs with the V2315CF will convert between the format of the disk files used on the simulator and the format on the microSD card that sits in a Mini 2315 Cartridge.

Another utility programs list all the V2315CF format disk image files in some selected folder, also listing the cartridge identifier and a description alongside the file name.

The selection of real or virtual mode for the V2315CF is done by changing a tiny switch on the 2310 Interface Board. Virtual mode will operate just as real mode does, but the 1130's disk drive is not spinning or making sounds.

Virtual 2315 Cartridge Facility

The SPI link exchanges 16 bit words between FPGA and Pico. The Pico is the master and always initiates each message. The first half of the word is a register address and the second half is used to transfer data.

Register 0x00 sets or clears cartridge ready status in the FPGA.

Register 0xA0 fetches that status plus other conditions from FPGA to Pico - file ready, real/virtual mode, unlocked, read-only, and fault latch.

Register 0x04 toggles the read-only state. Sent by the Pico when the Read Only toggle switch is momentarily activated.

Register 0x05 sends 1/3 of the SDRAM memory address in each message, thus three in a row are needed to select an address

Register 0x06 writes a word to SDRAM at the current address. Each message sends 1/2 of a word, so a pair of messages are needed for each word, after which the SDRAM address is auto-incremented.

Register 0x88 fetches a word from SDRAM at the then current address. Two messages are needed for each word, after the second the address is autoincremented.

Messages 0x05, 0x08 and 0x88 are the way the Pico will load a virtual cartridge image into SDRAM and fetch back the possibly changed contents to rewrite the image file during unload.

Virtual 2315 Cartridge Facility

Register 0x81 returns the current cylinder number from the FPGA to the Pico.

Register 0x82 returns the current sector number and head, the ready condition of the disk, and an ID indicating whether we are currently executing a seek, read or write.

Register 0x90 and 0x91 return the version number of the code running in the FPGA, so the Pico can verify that it is talking to a compatible version.

The Pico drives a serial link over the debug port that emits many diagnostic messages during startup and operation. It operates at 460,800 baud.

Hooking up a USB serial cable to the debug pins on the rear of the main V2315CF box and opening a terminal program at 460800 baud will display all the diagnostic messages as they are emitted.

Typing L into the terminal turns on logging. For every seek, read or write, a message will be printed with the cylinder number and for read or write, the head and sector as well. Typing S stops the logging.

Virtual 2315 Cartridge Facility

Key signals that are passed through the V2315CF between the disk controller and the disk drive are directly connected on the 2310 Interface Board using electronic switches when the unit is in real mode. This eliminates any delays having to read in a signal from one side and then emit it to the other side.

The output drivers for signals are open collector, thus in real mode those key signal output drives are driven by the FPGA to their high impedance open state. In virtual mode, these drive the signals from V2315CF to the 1130 and the electronic switch is open.

The FPGA is programmed in Verilog. As George Wiley recommended, I performed all simulation on Xilinx Vivado while synthesis and bitstream generate was performed on Lattice IceCube2. The bitstream is loaded to the flash rom over a JTAG connector on the board using Lattice Diamond programmer.

The Pico is programmed in C++. I use Visual Studio Code with the Pico support extension to set up the Pico SDK and build environment. The executable produced is loaded by booting the Pico as a USB drive on a PC then copying the file onto it.

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This project was developed to be shared with others working with IBM 1130 systems. Those who wish to evolve this to support other vintage computing systems are encouraged to do so, subject only to the restrictions below.

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